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Zeroth law of thermodynamics. If two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other. Two systems in thermal equilibrium have the same temperature. Temperature is what a thermometer measures. Heat is energy that flows from a hotter body to a colder one.

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First law of thermodynamics. Energy cannot be created or destroyed, only transferred or converted. The change in internal energy equals the heat added minus the work done by the system. An isolated system exchanges neither matter nor energy with its surroundings. A closed system exchanges energy but not matter. An open system exchanges both matter and energy.

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Second law of thermodynamics. The entropy of an isolated system never decreases. Heat flows spontaneously from hot to cold, never the other way without work. A reversible process is an idealisation in which the entropy stays constant. Third law of thermodynamics. As temperature approaches absolute zero, the entropy of a perfect crystal approaches zero. Absolute zero is 0 kelvin, or minus 273.15 degrees Celsius, and cannot be reached in a finite number of steps.

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Entropy. Entropy is a measure of the disorder of a system, or the number of microscopic arrangements that match its overall state. Ice melting increases entropy. A gas expanding into a vacuum also increases entropy. Entropy is measured in joules per kelvin.

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Heat engines. A heat engine turns heat into work by moving heat from a hot reservoir to a cold reservoir. Car engines and power stations are heat engines. Efficiency is the work done divided by the heat taken from the hot reservoir. Some heat must be rejected to the cold reservoir, as required by the second law, so no heat engine can be 100% efficient.

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Carnot efficiency. Carnot efficiency is the maximum efficiency of a heat engine between two reservoirs: 1 - Tc/Th, where Tc is the cold temperature and Th is the hot temperature. All temperatures must be in kelvin. Real engines always fall short of this limit. A refrigerator is a heat engine run backwards: it uses work to move heat from cold to hot.